RESEARCH STATEMENT: THEODORE WEISMAN

Introduction

My work focuses on the interaction between **discrete subgroups of Lie groups**, **geometric group theory**, and **geometric structures on manifolds**. The interplay between these areas can be traced back at least to 19th-century foundational work of Klein, who promoted the idea that different types of geometric spaces ought to be studied using their *symmetry groups*. Frequently, I work with *homogeneous spaces*, which have a high degree of symmetry at all scales and have important connections to low-dimensional topology, dynamics, and physics. Symmetries of homogeneous spaces are often studied using tools from Lie theory and Riemannian geometry, but another approach is to consider their "large-scale" properties (i.e. their *coarse geometry*). Historically, homogeneous and coarse geometry were closely linked via the study of group actions on "negatively curved" ("hyperbolic") spaces, inspiring revolutionary advances in both geometric topology and group theory. In the second half of the 20th century, work of Thurston and others made hyperbolic geometry fundamental to the modern understanding of three-dimensional manifolds; at around the same time, Gromov gave a far more general way to formulate the notion of "negative curvature," allowing ideas from smooth geometry to be applied to abstract countably infinite groups. This "coarse" perspective in turn became important in the original (smooth) setting. In the last 15 years, it was notably used to help resolve some of the foundational questions in 3-manifold theory (for example as a component of the proofs of the virtual Haken and virtual fibering conjectures, completed by Agol and Wise).

In both homogeneous and coarse geometry, relaxing the hyperbolicity assumption even slightly tends to introduce a much wider range of possible phenomena, opening up entire new avenues of research. For example, while discrete subgroups of $\mathrm{PSL}_2(\mathbb{C})$ (*Kleinian groups*) have a well-developed theory coming from their connection to hyperbolic geometry, almost nothing is known about the general behavior of discrete subgroups of *higher-rank* Lie groups (such as $\mathrm{SL}_d(\mathbb{R})$ and $\mathrm{SL}_d(\mathbb{C})$ for $d > 2$). Mathematicians are not remotely close to a classification of such groups; we do not even have an idea of what such a classification might look like. However, the possibilities are very exciting to explore; what is known is that these groups serve as symmetries of objects with rich geometric and dynamical structure coming from their non-hyperbolic features (e.g. *flat subspaces*).

Much of my research focuses on studying groups whose coarse geometric properties interact well with this structure, since it turns out that the "large-scale" perspective is powerful even in the presence of non-hyperbolic behavior. Moreover, the interplay actually goes both ways: my research program also applies ideas from higher-rank Lie groups and topological dynamics towards topics in geometric group theory, such as mapping class groups of surfaces. My philosophy is that the common ancestry between smooth and coarse geometry means that considering questions in one area yields insights which can be applied to the other, and this principle persists even though the specific techniques vary wildly between disciplines.

Research areas

Anosov subgroups and generalizations. Anosov subgroups are the "most hyperbolic" discrete subgroups of higher-rank Lie groups, generalizing the notion of a *convex cocompact* subgroup in rank one. They have many interesting properties, but I am mainly interested in using them as an entry point for understanding the larger (and more mysterious) class of discrete subgroups of Lie groups with some form of "hyperbolic behavior." Exploring this class involves a synthesis of theory-building and example-construction; the goal is to find good organizing principles for large families of discrete groups (Anosov and non-Anosov alike).

Topological dynamics in geometric group theory. One can often understand symmetries of geometric spaces by analyzing the dynamics of their action on a purely topological "boundary at infinity," and this turns out to be a remarkably powerful tool in a wide array of situations. This arm of my research program is unified more by its methods and philosophy than by its basic objects of study, and has led to collaboration with researchers working in a number of different subfields, including geometric group theory, low-dimensional

topology, and rigidity theory. By focusing on basic properties of boundary actions, and exploring the way they manifest geometric information, I am able to expose otherwise hard-to-spot connections between groups appearing in each of these areas.

Convex projective structures. Convex projective geometry is a far-reaching generalization of hyperbolic geometry, with important ties to both Anosov representations and *higher Teichmüller theory*. I am interested in convex projective structures because they can be studied simultaneously using techniques from coarse geometry, dynamics, and higher-rank symmetric spaces; I work to establish concrete connections between these areas as a way to understand the vast array of examples provided by the theory.

1. Anosov subgroups and generalizations

A central goal of homogeneous geometry is to understand the nature of discrete subgroups of Lie groups. These groups can display an overwhelming array of different behavior; to get a foothold, one is led to study families of discrete groups that are "nicely embedded." In hyperbolic geometry, *geometrically finite* groups play this role. Geometrically finite subgroups of $\mathrm{PSL}_2(\mathbb{C})$ have many strong algebraic and geometric properties, and can be characterized in terms of topology, dynamics, or coarse geometry [21]; they are also now known to be dense among finitely-generated Kleinian groups [77, 80]. Thus it is natural to ask:

Question 1. *What are the higher-rank analogs of geometrical finiteness? How are examples similar to their counterparts in hyperbolic space? More interestingly, how are they fundamentally different?*

In a seminal paper [63], Labourie took an important step forward by introducing the notion of an *Anosov subgroup* of $\mathrm{SL}_d(\mathbb{R})$. The definition (later expanded by Guichard–Wienhard [47]) was inspired by the dynamical behavior of *convex cocompact* groups, which make up the nicest subclass of geometrically finite groups. Like convex cocompact groups, Anosov subgroups are always abstractly Gromov-hyperbolic [57, 17]; they are also *stable*, in the sense that any small deformation of an Anosov subgroup remains Anosov. This last property ties them closely to the field of *higher Teichmüller theory* [98].

Anosov representations have attracted a great deal of attention since their introduction (see e.g. [45, 57, 17, 32, 58, 25]), but for many years, it was far from clear whether or not the existence of an Anosov embedding into $\mathrm{SL}_d(\mathbb{R})$ for a given hyperbolic group should be a typical property, or an unusual one. My collaborators and I made major progress by proving:

Theorem 2 (Douba–Flechélles–Weisman–Zhu [33])**.** *Every cubulated hyperbolic group admits an Anosov representation.*

Cubulated groups are "abundant," both in a precise sense coming from the theory of random groups [43, 81] and in the sense that there are numerous examples to be found in the literature [79, 99, 81, 11, 101, 41, 64], so our result has helped shape the general impression that "many" hyperbolic groups ought to have Anosov representations. It also allows for the construction of many examples with surprising properties. For example, an open question of Kapovich [54] asks about the existence of higher-dimensional Kleinian groups with certain "exotic" limit sets, and our theorem establishes that such limit sets can occur for Anosov subgroups. An open question of Wise [100] also asks if every cubulated hyperbolic group admits a convex cocompact representation in rank one; in a sense, our theorem gives an affirmative answer to the natural higher-rank version of this question.

The starting point of our proof of Theorem 2 is celebrated work of Agol [3] and Haglund–Wise [49] relating cubulated groups to *right-angled Coxeter groups*. Right-angled Coxeter groups can be understood using combinatorial geometry, but they also have a robust theory of linear representations dating back to work of Tits and Vinberg [19, 92]. Ultimately we construct the Anosov representations of Theorem 2 by connecting these two viewpoints, using a blend of techniques from CAT(0) cube complexes, projective geometry, and higher-rank symmetric spaces.

However, since Anosov subgroups are always hyperbolic, they are much too limited to give a full picture of the "geometrically interesting" discrete subgroups of higher-rank Lie groups; this is especially true given that there are many constructions of non-hyperbolic discrete groups coming from *convex projective geometry* (see Section 3 below). Nevertheless, the proof of Theorem 2 suggests ways to progress further. For example, one specific natural question I plan to explore is:

Question 3. *Do Tits–Vinberg representations restrict to structurally stable representations on arbitrary (non-hyperbolic) quasi-convex subgroups of Coxeter groups?*

A discrete subgroup of $\mathrm{SL}_d(\mathbb{R})$ is *structurally stable* if all of its small deformations are discrete and abstractly isomorphic to the original group. Classical work of Sullivan [88] shows that in $\mathrm{PSL}_2(\mathbb{C})$, structural stability is characteristic of either convex cocompactness or rigidity, but it is far from clear to what an extent an analogous statement holds in other Lie groups; Anosov representations are structurally stable [63, 47], but so are a number of other non-hyperbolic discrete groups (see e.g. [32]). A good answer to Question 3 could lead to a wholly new understanding of stability in higher rank, so I really view it as a concrete step in the much larger program of mapping out the geometry and dynamics of "hyperbolic-like" discrete groups. On a smaller scale, an interesting special case of Question 3 is its restriction to *relatively hyperbolic* quasi-convex subgroups of (right-angled) Coxeter groups; one concrete problem to start with (possibly suitable for a graduate student) would be to state and prove a relative version of Theorem 2.

Extended geometrically finite representations. In hyperbolic space, geometrically finite groups are precisely those whose distortion is confined to certain isolated regions. So, in higher rank, a natural way to move beyond the theory of Anosov representations is to consider subgroups with isolated "non-Anosov" behavior. There are many interesting examples of groups fitting this description, but a unified understanding had been missing. As part of my thesis work, I introduced *extended geometrically finite* (*EGF*) subgroups, which capture essentially all of the previously incompatible "relatively Anosov" phenomena:

Theorem 4 (Weisman [94, 97])**.** *The following are all EGF subgroups:*

- *The relatively Anosov subgroups defined by Kapovich–Leeb [56] and Zhu [103],*
- *Relatively hyperbolic groups in* $\mathrm{PGL}_{d+1}(\mathbb{R})$ *that are* $\mathbb{R}\mathrm{P}^d$-convex cocompact *in the sense of [32],*
- *Holonomy groups of many convex projective manifolds with* generalized cusps *(see [29]),*
- *Limits of Anosov triangle reflection groups in* $\mathrm{PGL}_3(\mathbb{R})$ *(see [66]),*
- *New examples not previously considered by other authors.*

The definition of an EGF representation is rooted in topological dynamics. The key insight is that one can relativize the definition of an Anosov representation into a semisimple Lie group G by embedding a "blowup" of the ideal boundary of a relatively hyperbolic group into a flag manifold associated to G. Allowing nontrivial blowups lets the theory encompass the wide range of examples mentioned above; however, the definition is also strong enough to support a powerful relative stability theorem:

Theorem 5 (Weisman [94])**.** *If ρ is an EGF representation, and ρ' is a small deformation of ρ that satisfies a peripheral condition, then ρ' is EGF.*

A special case of this result was proved independently (using different methods) by Zhu–Zimmer [104]. However, in addition to its greater generality, another significant advantage of Theorem 5 is that the peripheral condition in this version of the theorem is much more flexible: it allows for deformations which may not even be topologically conjugate to the original, and it can sometimes even be dropped entirely (something I explored in [97]).

Since their introduction, EGF representations have attracted attention from other researchers [93, 15], and they have many potential applications. For instance, there are several interesting open questions about the nature of boundaries of spaces of Anosov representations [65], and evidence shows that EGF representations make up a portion of these boundaries [66]; I have plans with Maloni to use some the *combination theorems* discussed in the next section to show how to use EGF representations to explore the boundary of the space of *quasi-Hitchin representations*. As another example, I expect that EGF subgroups could be combined with techniques in my work with Tsouvalas [90] on combination theorems in $\mathrm{SL}_d(\mathbb{R})$ to prove a "relativization theorem" for quasi-isometrically embedded discrete subgroups.

In recent work, I also used EGF representations as a key piece in the proof of a far-reaching generalization of (one part of) Thurston's famous *hyperbolic Dehn filling theorem*. Thurston's theorem shows that one can find hyperbolic structures on *Dehn fillings* of a noncompact finite-volume hyperbolic 3-manifold M by applying a small deformation to the holonomy group of M in $\mathrm{PSL}_2(\mathbb{C})$. I proved:

Theorem 6 (Weisman [95])**.** *There is a general notion of "Dehn filling" (generalizing Dehn filling in $\mathbb{H}^3$) for an EGF subgroup Γ in any semisimple Lie group G. When Dehn filling occurs, Γ is realized as a limit of pairwise non-isomorphic discrete groups Γ_n converging to Γ in a geometric sense.*

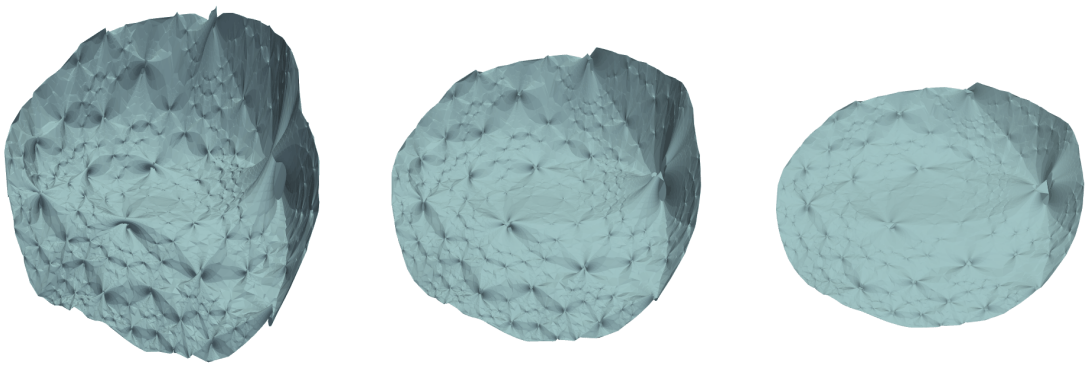


FIGURE 1. Limit sets for a family of convex cocompact subgroups in $\mathrm{SU}(3,1)$. Each limit set is homeomorphic to a *Pontryagin sphere* (see Figure 2 below) embedded in a 5-sphere, so the pictures depict projections to a 3-dimensional slice.

The proof of this theorem further develops some technology originally built for the proof of Theorem 5, combining it with a coarse geometric analog of hyperbolic Dehn surgery: the *relatively hyperbolic Dehn filling theorem* of Osin [82] and Groves–Manning [44]. Theorem 6 provides a common framework for understanding a number of different examples of "Dehn filling" in homogeneous spaces [86, 1, 2, 27, 6], and should lead directly to brand-new examples as well—in many cases with surprising behavior. For instance, my upcoming work with Danciger applies Theorem 6 to prove the following:

Theorem 7 (Danciger–Weisman, in progress)**.** *Let Γ be the fundamental group of a finite-volume hyperbolic 3-manifold M with one cusp, satisfying the IPRB condition. Then the $\mathrm{SU}(3,1)$-character variety of Γ contains a countably infinite family of "hyperbolic Dehn filling spaces."*

The behavior described by Theorem 7 has no Kleinian analog, and is very different from what has previously been observed in other contexts. For example, one remarkable consequence of the theorem is that, for convex cocompact subgroups of $\mathrm{SU}(3,1)$, the breakdown between global rigidity and local rigidity can be arbitrarily large:

Corollary 8 (Danciger–Weisman, in progress)**.** *Given any $n \in \mathbb{N}$, there exists a closed hyperbolic 3-manifold M and a collection $\{\Gamma_i\}_{i=1}^n$ of locally rigid convex cocompact subgroups of $\mathrm{SU}(3,1)$ such that each Γ_i is isomorphic to $\pi_1 M$, but the Γ_i are pairwise non-conjugate in $\mathrm{SU}(3,1)$.*

Danciger and I are also able to use the Dehn filling approach to produce examples of convex cocompact subgroups of $\mathrm{SU}(3,1)$ with "exotic" limit sets (see Figure 1):

Theorem 9 (Danciger–Weisman, in progress)**.** *There exist convex cocompact subgroups of $\mathrm{SU}(3,1)$ whose limit set in $\partial\mathbb{H}^3_{\mathbb{C}}$ is homeomorphic to a* Pontryagin sphere*.*

Groups with Pontryagin sphere limit set have also recently been found (using a wholly different construction) in $\mathrm{SO}(4,1)$ [34]; it would be interesting to explore whether our construction can be made to work in real hyperbolic spaces as well. Theorem 6 also immediately suggests the following question:

Question 10. *What further examples of "Dehn filling" phenomena can occur in* higher rank *Lie groups?*

Like many of my proposed research questions, this is (deliberately) vague, since it is hard to know what sort of behavior to expect; one could start by looking for concrete answers in specific low-dimensional Lie groups (for a number of reasons, $\mathrm{SO}(3,2)$ is a natural candidate).

2. Topological dynamics in group theory and topology

Any hyperbolic metric space X has an associated "boundary at infinity" ∂X, where isometries of X act by homeomorphisms. Amazingly, it is often possible to recover essentially all of the geometry of X by analyzing the induced action of its isometry group on this ideal boundary [22, 23, 102]. So, if one wants to understand the "hyperbolic" nature of some arbitrary metric space Y, a good strategy is to study the way that the

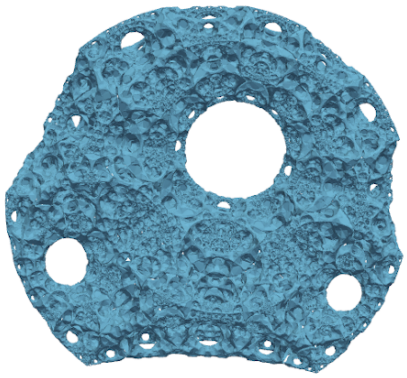


FIGURE 2. A Pontryagin sphere limit set for a discrete subgroup of $\mathrm{SO}(4,1)$, discovered recently by Douba–Lee–Marquis–Ruffoni [34].

dynamics of various "actions at infinity" for $\mathrm{Isom}(Y)$ inform the geometry of Y. At the most basic level, one can ask whether or not the action of $\mathrm{Isom}(Y)$ on ∂Y resembles a *convergence action*, which formalizes the "north–south dynamics" of hyperbolic isometries. Although convergence actions have a simple definition, they capture suprisingly detailed information; understanding "convergence-like" behavior has led me to both new results in the geometry and dynamics of hyperbolic groups, and to deeper understanding of phenomena occuring in fields that might not otherwise have an obvious connection.

Topological stability. When a group Γ acts by homeomorphisms on a space M, it is often natural to ask what features of the action are preserved when one applies a small perturbation in the space $\mathrm{Hom}(\Gamma, \mathrm{Homeo}(M))$; when the original action on M has a geometric interpretation, this approach can yield rigidity and stability results for classes of geometric structures. This basic idea forms an important part of the history of rigidity theory [87, 40, 52, 59], and some of my work shows how it can play a role in more modern contexts as well.

For example, as part of my thesis work, I recontextualized an old technique of Sullivan [88] to show that one can "encode" actions of certain discrete subgroups of Lie groups on *flag manifolds* (such as real projective space). This led to the construction of a combinatorial object I called a *quasi-geodesic automaton*, which formed the basis of the proof of the stability theorem for EGF representations (Theorem 5 above). Mann, Manning, and I showed that this construction could actually also be used as the starting point for a stability result in a different context:

Theorem 11 (Mann–Manning–Weisman [71])**.** *If Γ is any hyperbolic group, then the natural action of Γ on its boundary is topologically stable.*

Here, *topological stability* means that any sufficiently small C^0 perturbation of the action is "semi-conjugate" to the original action; semi-conjugacy is the appropriate notion of equivalence in the C^0 setting, so the theorem can also be interpreted as a rigidity result. Special cases of Theorem 11 had previously been shown using different methods by Bowden–Mann [20] and Mann–Manning [70]. Our techniques further extended to prove a relative version of the theorem:

Theorem 12 (Mann–Manning–Weisman [72])**.** *Suppose Γ is a* relatively *hyperbolic group. Then the natural action of Γ on its boundary $\partial\Gamma$ is* relatively *stable: any small perturbation of the action, subject to a condition on the peripheral actions, is semi-conjugate to the original.*

In more recent work, Nguyen and I showed that the methods used to prove the results above are also powerful enough to apply even in highly non-hyperbolic settings. We were motivated in part by recent progress on the *Zimmer program*, which seeks to understand and classify smooth actions of lattices in real Lie groups on low-dimensional manifolds. Generalizing earlier work of Kanai [52] and Katok–Spatzier [59], Brown–Rodriguez Hertz–Wang [24] recently proved a C^r-rigidity result for $\mathrm{SL}_d(\mathbb{R})$-lattice actions on flag varieties in $\mathbb{R}^d$; Kapovich–Kim–Lee [55] and Connell–Islan–Nguyen–Spatzier [28] respectively proved Lipschitz and C^0-stability results for uniform lattices. With different methods, Nguyen and I proved a non-Archimedean counterpart to these results:

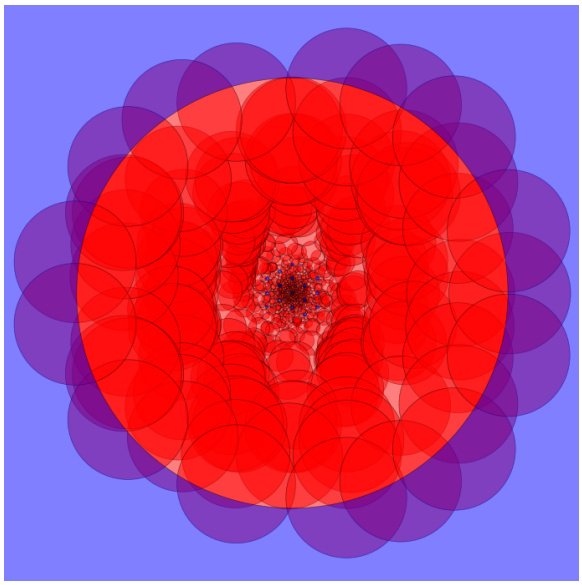


FIGURE 3. In some cases, it is possible to give an explicit construction for the *quasi-geodesic automaton* encoding "actions at infinity" for certain groups. This picture depicts part of the combinatorial data for an automaton encoding the action of the *figure-eight knot group* on $\mathbb{C}\mathrm{P}^1$, in the form of a covering of $\mathbb{C}\mathrm{P}^1$ by finitely many disks.

Theorem 13 (Nguyen–Weisman [78])**.** *Let G be a semisimple Lie group over a non-Archimedean local field, let $Q < G$ be a parabolic subgroup, and let Γ be a lattice in G. Then the action $\Gamma \to \mathrm{Homeo}(G/Q)$ is topologically stable.*

Our proof was inspired by the techniques developed for Theorem 11 and Theorem 12, and by the notion of "meandering–hyperbolicity" introduced in work of Kapovich–Kim–Lee [55]. By blending the ideas, we developed a combinatorial connection between the homeomorphic action of Γ on the space G/Q and the isometric action of Γ on the *Euclidean building* associated to G. Euclidean buildings are typically non-homogeneous spaces, and like higher-rank symmetric spaces, they have abundant flat subspaces interacting with each other in a complicated manner. Our proof of Theorem 13 shows that, armed with the right perspective, one can still apply hyperbolic ideas to explore the asymptotic properties of isometric actions in this context as well.

Geometric finiteness in mapping class groups. It has long been known that there is an imperfect but highly fruitful analogy between hyperbolic geometry and the geometry of the *Teichmüller space* of a closed hyperbolic surface Σ [12, 73, 74, 83]. Since the *mapping class group* $\mathrm{MCG}(\Sigma)$ of Σ acts properly on Teichmüller space, the analogy also endows it with subtle but fundamental hyperbolicity properties [74, 75].

It is natural to also ask about the hyperbolicity properties of a *subgroup* $\Gamma < \mathrm{MCG}(\Sigma)$, especially because there is a correspondence between such subgroups and isomorphism classes of surface bundles [85]. Building on a foundational result of Thurston, Farb–Mosher [38] and Hamenstädt [50] proved that the "most hyperbolic" subgroups of $\mathrm{MCG}(\Sigma)$ (the *convex cocompact* subgroups) correspond precisely to surface bundles with hyperbolic fundamental group. There are now many characterizations of convex cocompactness in $\mathrm{MCG}(\Sigma)$ (see e.g. [61, 62, 38, 13]); many of the definitions are inspired by analogies with Kleinian groups, and draw from the rich interplay between the geometry of Teichmüller space, the dynamics of the $\mathrm{MCG}(\Sigma)$-action on the space of *measured laminations* on Σ, and the geometry of the *complex of curves*.

In [76], Mosher asked for a theory of *geometric finiteness* for subgroups of $\mathrm{MCG}(\Sigma)$, hoping to gain a more general understanding of the interaction between surface bundles and the geometry and dynamics of mapping class groups. Mosher was partly motivated by the fact that, while convex cocompact subgroups of $\mathrm{MCG}(\Sigma)$ can be hard to find (although see recent breakthrough work of Kent–Leininger [60]), there are a number of available constructions for subgroups that are "almost" convex cocompact, away from some isolated exceptional behavior [67, 91, 69]. However, work of Leininger [68] showed that Mosher's initial suggestions

for a definition of "geometric finiteness" failed to capture many of these natural examples. While there has been recent progress in the area [35, 36, 18, 91, 4], we are still missing a theory of geometric finiteness in $\mathrm{MCG}(\Sigma)$ that both incorporates a wide array of examples and ties together the geometric, topological, and dynamical pictures in a satisfying way.

In my upcoming work with Dowdall and Maloni, we propose a way to fill this gap with a wholly new class of subgroups; we discovered that there is actually a way to rethink the Farb–Mosher and Kent–Leininger definitions of convex cocompactness that allows for a comprehensive "relativized" version of the theory. The simplest way to identify a subgroup Γ in our class is to ask whether "lined up" sequences in Γ (i.e., sequences lying in a neighborhood of a geodesic in Γ) are sent, under an orbit map into Teichmüller space, to sequences that are "lined up" along a Teichmüller geodesic. We call such subgroups *relatively aligned*, and prove that there are many different ways to interpret this property:

Theorem 14 (Dowdall–Maloni–Weisman, in preparation)**.** *Let Γ be a subgroup of $\mathrm{MCG}(\Sigma)$. The following are equivalent:*

(1) Γ is relatively aligned;
(2a) The identity map on Γ extends continuously to a well-behaved boundary map on the limit set of Γ in the Thurston boundary of Teichmüller space.
(2b) The identity map on Γ extends continuously to a canonical compactification of Γ in $\mathrm{MCG}(\Sigma)$.
(3) Γ is relatively quasi-isometrically embedded into the curve graph of Σ and has transitively intersecting peripherals.

Our proof of Theorem 14 introduces new techniques beyond those that have been previously used to study convex cocompact groups in $\mathrm{MCG}(\Sigma)$. One of our key insights was that, since analogies between Teichmüller space and hyperbolic space can only be pushed so far, there is a great deal to be gained by borrowing tools and ideas that originated in the study of higher-rank Lie groups. For example, the dynamical versions of our definition ((2a) and (2b) above) were inspired by the "blowup" behavior of the EGF representations discussed in the previous section. Further, to prove many of the equivalences between our different definitions, we rely heavily on a new compactification of the mapping class group (alluded to in (2b)), which is analogous to the *wonderful compactification* of a semisimple Lie group. These new tools are powerful enough to prove strong general results, unrelated to geometric finiteness. We show:

Theorem 15 (Dowdall–Maloni–Weisman, in preparation)**.** *Let f be an arbitrary element of* MCG*. Then, for any $\varepsilon > 0$, there is a lower bound on the length of f in the curve graph which is uniformly linear in the "ε-thick length" of f in Teichmüller space.*

This strengthens an earlier result of Rafi [83], but our approach (which centers topological dynamics) is completely new; remarkably, the proof of Theorem 15 does not even rely on subsurface projections.

Combination theorems. One interesting consequence of my work with Dowdall and Maloni is that there are striking similarities between EGF subgroups of higher rank Lie groups and many subgroups of mapping class groups. Maloni and I are working to formalize this observation through a unified framework of *extended convergence actions*; the advantage of doing so is that one can prove theorems in both contexts simultaneously. For example, Maloni and I plan to generalize a *combination theorem* I proved with Traaseth, in the spirit of Klein and Maskit's classical results for Kleinian groups:

Theorem 16 (Traaseth–Weisman [89])**.** *There are "ping-pong" criteria guaranteeing that an amalgamated free product or HNN extension of geometrically finite convergence groups is another geometrically finite convergence group.*

With our framework in place, Maloni and I will be able to prove a generalization of Theorem 16 to the setting of *extended* convergence groups. This will yield a simultaneous proof of a procedure for constructing families of examples of both EGF subgroups and "geometrically finite" subgroups of $\mathrm{MCG}(\Sigma)$. Both specializations of the general theorem should be new results in their respective contexts, and will give verifiable criteria useful for explicit constructions.

3. Convex projective geometry

Another major branch of my research program concerns *convex projective structures*. Any convex open subset $\Omega \subset \mathbb{R}\mathrm{P}^d$ can be equipped with its *Hilbert metric* d_Ω, which is invariant under the group $\mathrm{Aut}(\Omega) <$

$\mathrm{PGL}_{d+1}(\mathbb{R})$ of projective transformations preserving Ω. When Ω is a round ball, this recovers the definition of the *projective model* of d-dimensional hyperbolic space, but the general family of metric spaces that can arise this way is much broader. In particular, *any* discrete subgroup of a linear Lie group can be made to act properly by isometries on a convex domain in $\mathbb{R}\mathrm{P}^d$; thus the Hilbert metric (and other features of the projective geometry) make it possible to directly adapt arguments from hyperbolic space towards the study of an enormous family of interesting groups. Consequently convex projective geometry has developed close connections with Anosov representations [32, 105] and higher Teichmüller theory [42, 26, 48, 9].

The hyperbolic tools are most useful in this context if one assumes that the action on Ω is especially well-behaved (say, cocompact). Examples of this sort tend to display an intriguing mix of "hyperbolic" and "higher-rank" behavior [10, 53, 5, 27, 15, 31], but they are far from fully understood. I am interested in exploring aspects of these examples from several different perspectives. For example, with Islam [51], I initiated a general investigation of the interplay between the coarse geometry of a discrete group Γ acting cocompactly on Ω and the geometry of the embedding $\Gamma \hookrightarrow \mathrm{SL}_d(\mathbb{R})$. We focused on studying "hyperbolic directions" in the group Γ. There are (at least) three ways to make this precise:

(i) the *coarse geometric* sense: a geodesic in Γ is "hyperbolic" if it satisfies the *Morse property* when Γ is viewed as an abstract finitely-generated group.
(ii) the *Lie-theoretic* sense: a geodesic in $\Gamma < \mathrm{SL}_{d+1}(\mathbb{R})$ is "hyperbolic" if it is *uniformly regular*, meaning it behaves like a geodesic in an Anosov subgroup of $\mathrm{SL}_{d+1}(\mathbb{R})$.
(iii) the *projective geometric* sense: a geodesic in Γ is "hyperbolic" if it corresponds to a path in Ω whose endpoint in the hypersurface $\partial\Omega$ is a C^α-regular point for some $\alpha > 1$.

Foundational results of Benoist [8] and Guichard [46] can be reinterpreted as saying that if *every* geodesic in Γ satisfies one of (i), (ii), or (iii), then every geodesic also satisfies both of the other two; in these cases, Γ is "as hyperbolic as possible." Islam and I showed that the different notions of hyperbolicity still correspond for an *arbitrary* (not necessarily hyperbolic) group Γ:

Theorem 17 (Islam–Weisman [51])**.** *If $\Gamma < \mathrm{SL}_{d+1}(\mathbb{R})$ is a discrete group acting cocompactly on a convex domain $\Omega \subset \mathbb{R}\mathrm{P}^d$, then there is an exact correspondence between the geodesics in Γ satisfying (i), (ii), and (iii) above.*

As a byproduct of the proof, Islam and I also established that the metric space $(\Omega, \mathrm{d}_\Omega)$ satisfies the *Morse local-to-global* property recently introduced by Russell–Spriano–Tran [84]; this shows that convex domains fit into a nice quasi-isometry invariant class of metric spaces with several desirable properties [37].

Convex cocompactness in $\mathbb{R}\mathrm{P}^d$. In [32], Danciger–Guéritaud–Kassel introduced a general notion of *convex cocompactness* for group actions on convex domains in $\mathbb{R}\mathrm{P}^d$. Groups satisfying this definition are called *projectively convex cocompact*, and have many desirable geometric properties. For example, they are always quasi-isometrically embedded into $\mathrm{SL}_{d+1}(\mathbb{R})$ and are stable under small deformations. The theory is partly motivated by a close connection to Anosov representations: there is a strong (essentially one-to-one) correspondence between Anosov subgroups and projectively convex cocompact groups that are also abstractly word-hyperbolic. However, the definition also provides a broad generalization of Anosov representations, since it admits a wide array of non-hyperbolic examples.

The definition of a projectively convex cocompact subgroup is based in geometric topology, but as part of my thesis work, I showed that there is also a dynamical interpretation:

Theorem 18 (Weisman [96])**.** *Let Γ be a discrete subgroup of $\mathrm{PGL}_{d+1}(\mathbb{R})$ preserving a properly convex open set $\Omega \subset \mathbb{R}\mathrm{P}^d$. Then Γ is convex cocompact if and only if there is a closed "boundary-convex" subset $\Lambda \subseteq \partial\Omega$ so that Γ is* expanding at the faces *of Λ.*

Part of the appeal of Anosov representations comes from their large number of different characterizations, each with a different feel and each highlighting a different aspect of the general theory. Theorem 18 generalizes an "expansion/contraction" characterization due to Kapovich–Leeb–Porti [57], thus enriching the theory of general projectively convex cocompact groups. To prove the theorem, I developed several new projective-geometric tools, including an effective version of the *Benzécri cocompactness theorem*. Theorem 18 itself was instrumental in my proof of the fact that any relatively hyperbolic projectively convex cocompact subgroup is extended geometrically finite (see Theorem 4 above).

Geometrical finiteness in $\mathbb{R}\mathrm{P}^d$. A natural next step for further research would be to consider notions of *geometrical finiteness* for group actions on convex projective domains. Preliminary work in this direction was completed by Blayac–Crampon–Marquis [30, 14], who developed several notions of geometrical finiteness for actions on *strictly* convex projective domains. However, none of these definitions are fully compatible with either the projectively convex cocompact subgroups considered by Danciger–Guéritaud–Kassel or a number of other natural examples [7, 16, 39]. In many cases, the groups in question are extended geometrically finite (see above), but this only provides an interpretation of geometrical finiteness "at infinity." Thus one is led to consider:

Problem. *Develop a characterization of EGF actions on a convex domain Ω in $\mathbb{R}\mathrm{P}^d$ in terms of the geometric topology of the quotient space.*

On the other hand, it is also possible to construct examples of projectively convex cocompact representations that are *not* EGF; the subgroups in question are isomorphic to uniform lattices in some $\mathrm{SL}_d(\mathbb{R})$. So, a more ambitious goal would be to develop a unified notion of "geometrical finiteness" capturing all of the relevant examples. It is far from clear if such a definition should even exist, but there are definite parallels between the different types of examples that can appear, and further investigation should clarify the situation greatly.

References

[1] Miguel Acosta. "Spherical CR Dehn surgeries". In: *Pacific J. Math.* 284.2 (2016), pp. 257–282.

[2] Miguel Acosta. "Spherical CR uniformization of Dehn surgeries of the Whitehead link complement". In: *Geom. Topol.* 23.5 (2019), pp. 2593–2664.

[3] Ian Agol. "The virtual Haken conjecture". In: *Doc. Math.* 18 (2013). With an appendix by Agol, Daniel Groves, and Jason Manning, pp. 1045–1087.

[4] Tarik Aougab, Harrison Bray, Spencer Dowdall, Hannah Hoganson, Sara Maloni, and Brandis Whitfield. "Constructing reducibly geometrically finite subgroups of the mapping class group". In: *arXiv e-prints*, arXiv:2501.13234 (Jan. 2025), arXiv:2501.13234.

[5] Samuel Ballas, Jeffrey Danciger, and Gye-Seon Lee. "Convex projective structures on non-hyperbolic three-manifolds". In: *Geometry & Topology* 22 (Aug. 2015).

[6] Samuel Ballas, Jeffrey Danciger, Gye-Seon Lee, and Ludovic Marquis. *Exotic real projective Dehn filling spaces*. In preparation.

[7] Samuel A. Ballas. "Constructing convex projective 3-manifolds with generalized cusps". In: *J. Lond. Math. Soc. (2)* 103.4 (2021), pp. 1276–1313.

[8] Yves Benoist. "Convexes divisibles. I". In: *Algebraic groups and arithmetic*. Tata Inst. Fund. Res., Mumbai, 2004, pp. 339–374.

[9] Yves Benoist. "Convexes divisibles. III". In: *Ann. Sci. École Norm. Sup. (4)* 38.5 (2005), pp. 793–832.

[10] Yves Benoist. "Convexes divisibles. IV. Structure du bord en dimension 3". In: *Invent. Math.* 164.2 (2006), pp. 249–278.

[11] Nicolas Bergeron and Daniel T. Wise. "A boundary criterion for cubulation". In: *Amer. J. Math.* 134.3 (2012), pp. 843–859.

[12] Lipman Bers. "An extremal problem for quasiconformal mappings and a theorem by Thurston". In: *Acta Math.* 141.1-2 (1978), pp. 73–98.

[13] Mladen Bestvina, Kenneth Bromberg, Autumn E. Kent, and Christopher J. Leininger. "Undistorted purely pseudo-Anosov groups". In: *J. Reine Angew. Math.* 760 (2020), pp. 213–227.

[14] Pierre-Louis Blayac, Mickaël Crampon, and Ludovic Marquis. "Finitude géométrique en géométrie de Hilbert + an erratum/addendum". In: *arXiv e-prints*, arXiv:1202.5442 (Feb. 2012), arXiv:1202.5442.

[15] Pierre-Louis Blayac and Gabriele Viaggi. "Divisible convex sets with properly embedded cones". In: *Publ. Math. Inst. Hautes Études Sci.* 141 (2025), pp. 99–189.

[16] M. Bobb. "Convex projective manifolds with a cusp of any non-diagonalizable type". In: *J. Lond. Math. Soc. (2)* 100.1 (2019), pp. 183–202.

[17] Jairo Bochi, Rafael Potrie, and Andrés Sambarino. "Anosov representations and dominated splittings". In: *J. Eur. Math. Soc. (JEMS)* 21.11 (2019), pp. 3343–3414.

[18] Eliot Bongiovanni. "Extensions of finitely generated Veech groups". In: *arXiv e-prints*, arXiv:2406.11090 (June 2024), arXiv:2406.11090.

[19] Nicolas Bourbaki. *Éléments de mathématique. Fasc. XXXIV. Groupes et algèbres de Lie. Chapitre IV: Groupes de Coxeter et systèmes de Tits. Chapitre V: Groupes engendrés par des réflexions. Chapitre VI: systèmes de racines.* Actualités Scientifiques et Industrielles [Current Scientific and Industrial Topics], No. 1337. Hermann, Paris, 1968, 288 pp. (loose errata).

[20] Jonathan Bowden and Kathryn Mann. "C^0 stability of boundary actions and inequivalent Anosov flows". In: *Ann. Sci. Éc. Norm. Supér. (4)* 55.4 (2022), pp. 1003–1046.

[21] B. H. Bowditch. "Geometrical finiteness for hyperbolic groups". In: *J. Funct. Anal.* 113.2 (1993), pp. 245–317.

[22] Brian H. Bowditch. "A topological characterisation of hyperbolic groups". In: *J. Amer. Math. Soc.* 11.3 (1998), pp. 643–667.

[23] Brian H. Bowditch. "Relatively hyperbolic groups". In: *Internat. J. Algebra Comput.* 22.3 (2012), pp. 1250016, 66.

[24] Aaron Brown, Federico Rodriguez Hertz, and Zhiren Wang. "Global smooth and topological rigidity of hyperbolic lattice actions". In: *Ann. of Math. (2)* 186.3 (2017), pp. 913–972.

[25] Richard Canary. *Anosov representations: Informal lecture notes.* 2021.

[26] Suhyoung Choi and William M Goldman. "Convex real projective structures on closed surfaces are closed". In: *Proceedings of the American Mathematical Society* 118.2 (1993), pp. 657–661.

[27] Suhyoung Choi, Gye-Seon Lee, and Ludovic Marquis. "Convex projective generalized Dehn filling". In: *Ann. Sci. Éc. Norm. Supér. (4)* 53.1 (2020), pp. 217–266.

[28] Chris Connell, Mitul Islam, Thang Nguyen, and Ralf Spatzier. *Boundary actions of lattices and C^0 local semi-rigidity.* Preprint, arXiv:2303.00543. 2023.

[29] Daryl Cooper, Darren Long, and Stephan Tillmann. "Deforming convex projective manifolds". In: *Geom. Topol.* 22.3 (2018), pp. 1349–1404.

[30] Mickaël Crampon and Ludovic Marquis. "Finitude géométrique en géométrie de Hilbert". In: *Ann. Inst. Fourier (Grenoble)* 64.6 (2014), pp. 2299–2377.

[31] Jeffrey Danciger, François Guéritaud, and Fanny Kassel. "Combination theorems in convex projective geometry". In: *arXiv e-prints*, arXiv:2407.09439 (July 2024), arXiv:2407.09439.

[32] Jeffrey Danciger, François Guéritaud, and Fanny Kassel. "Convex cocompact actions in real projective geometry". In: *arXiv e-prints*, arXiv:1704.08711 (Apr. 2017), arXiv:1704.08711.

[33] Sami Douba, Balthazar Fléchelles, Theodore Weisman, and Feng Zhu. "Cubulated hyperbolic groups admit Anosov representations". In: *Geom. Topol.* 29.9 (2025), pp. 4695–4766.

[34] Sami Douba, Gye-Seon Lee, Ludovic Marquis, and Lorenzo Ruffoni. "Convex cocompact groups in real hyperbolic spaces with limit set a Pontryagin sphere". In: *Bull. Lond. Math. Soc.* 58.4 (2026), Paper No. e70319, 18.

[35] Spencer Dowdall, Matthew G. Durham, Christopher J. Leininger, and Alessandro Sisto. "Extensions of Veech groups I: A hyperbolic action". In: *J. Topol.* 16.2 (2023), pp. 757–805.

[36] Spencer Dowdall, Matthew G. Durham, Christopher J. Leininger, and Alessandro Sisto. "Extensions of Veech groups II: Hierarchical hyperbolicity and quasi-isometric rigidity". In: *Comment. Math. Helv.* 99.1 (2024), pp. 149–228.

[37] Cornelia Drutu, Davide Spriano, and Stefanie Zbinden. "Weak Morse properties in spaces with bounded combings". In: *arXiv e-prints*, arXiv:2503.21325 (Mar. 2025), arXiv:2503.21325.

[38] Benson Farb and Lee Mosher. "Convex cocompact subgroups of mapping class groups". In: *Geom. Topol.* 6 (2002), pp. 91–152.

[39] Balthazar Fléchelles. "Geometric finiteness in convex projective geometry". Theses. Université Paris-Saclay, Sept. 2024.

[40] Étienne Ghys. "Rigidité différentiable des groupes fuchsiens". In: *Inst. Hautes Études Sci. Publ. Math.* 78 (1993), 163–185 (1994).

[41] Anne Giralt. "Cubulation of Gromov-Thurston manifolds". In: *Groups Geom. Dyn.* 11.2 (2017), pp. 393–414.

[42] William M Goldman. "Convex real projective structures on compact surfaces". In: *Journal of Differential Geometry* 31.3 (1990), pp. 791–845.

[43] Misha Gromov. "Asymptotic invariants of infinite groups". In: *Geometric group theory, Vol. 2 (Sussex, 1991)*. Vol. 182. London Math. Soc. Lecture Note Ser. Cambridge Univ. Press, Cambridge, 1993, pp. 1–295.

[44] Daniel Groves and Jason Fox Manning. "Dehn filling in relatively hyperbolic groups". In: *Israel J. Math.* 168 (2008), pp. 317–429.
[45] François Guéritaud, Olivier Guichard, Fanny Kassel, and Anna Wienhard. "Anosov representations and proper actions". In: *Geom. Topol.* 21.1 (2017), pp. 485–584.
[46] Olivier Guichard. "Sur la régularité Hölder des convexes divisibles". In: *Ergodic Theory Dynam. Systems* 25.6 (2005), pp. 1857–1880.
[47] Olivier Guichard and Anna Wienhard. "Anosov representations: domains of discontinuity and applications". In: *Invent. Math.* 190.2 (2012), pp. 357–438.
[48] Olivier Guichard and Anna Wienhard. "Convex foliated projective structures and the Hitchin component for $\mathrm{PSL}_4(\mathbf{R})$". In: *Duke Mathematical Journal* 144.3 (2008), pp. 381–445.
[49] Frédéric Haglund and Daniel T. Wise. "Special cube complexes". In: *Geom. Funct. Anal.* 17.5 (2008), pp. 1551–1620.
[50] Ursula Hamenstädt. "Word hyperbolic extensions of surface groups". In: *arXiv Mathematics e-prints*, math/0505244 (May 2005), math/0505244.
[51] Mitul Islam and Theodore Weisman. "Morse properties in convex projective geometry". In: *Adv. Math.* 479 (2025), Paper No. 110430.
[52] M. Kanai. "A new approach to the rigidity of discrete group actions". In: *Geom. Funct. Anal.* 6.6 (1996), pp. 943–1056.
[53] Michael Kapovich. "Convex projective structures on Gromov-Thurston manifolds". In: *Geom. Topol.* 11 (2007), pp. 1777–1830.
[54] Michael Kapovich. "Kleinian groups in higher dimensions". In: *Geometry and dynamics of groups and spaces*. Vol. 265. Progr. Math. Birkhäuser, Basel, 2008, pp. 487–564.
[55] Michael Kapovich, Sungwoon Kim, and Jaejeong Lee. "Structural stability of meandering-hyperbolic group actions". In: *J. Inst. Math. Jussieu* 23.2 (2024), pp. 753–810.
[56] Michael Kapovich and Bernhard Leeb. "Relativizing characterizations of Anosov subgroups, I". In: *Groups Geom. Dyn.* 17.3 (2023). With an appendix by Gregory A. Soifer, pp. 1005–1071.
[57] Michael Kapovich, Bernhard Leeb, and Joan Porti. "Anosov subgroups: dynamical and geometric characterizations". In: *Eur. J. Math.* 3.4 (2017), pp. 808–898.
[58] Fanny Kassel and Rafael Potrie. "Eigenvalue gaps for hyperbolic groups and semigroups". In: *J. Mod. Dyn.* 18 (2022), pp. 161–.
[59] A. Katok and R. J. Spatzier. "Differential rigidity of Anosov actions of higher rank abelian groups and algebraic lattice actions". In: *Tr. Mat. Inst. Steklova* 216.Din. Sist. i Smezhnye Vopr. (1997), pp. 292–319.
[60] Autumn E. Kent and Christopher J. Leininger. "Atoroidal surface bundles". In: *arXiv e-prints*, arXiv:2405.12067 (May 2024), arXiv:2405.12067.
[61] Autumn E. Kent and Christopher J. Leininger. "Shadows of mapping class groups: capturing convex cocompactness". In: *Geom. Funct. Anal.* 18.4 (2008), pp. 1270–1325.
[62] Autumn E. Kent and Christopher J. Leininger. "Uniform convergence in the mapping class group". In: *Ergodic Theory Dynam. Systems* 28.4 (2008), pp. 1177–1195.
[63] François Labourie. "Anosov flows, surface groups and curves in projective space". In: *Invent. Math.* 165.1 (2006), pp. 51–114.
[64] Jean-François Lafont and Lorenzo Ruffoni. "Special cubulation of strict hyperbolization". In: *arXiv preprint arXiv:2206.03620* (2022).
[65] Max Lahn. "Reducible Suspensions of Anosov Representations". In: *arXiv e-prints*, arXiv:2312.09886 (Dec. 2023), arXiv:2312.09886.
[66] Gye-Seon Lee, Jaejeong Lee, and Florian Stecker. "Anosov triangle reflection groups in SL(3,R)". In: *arXiv e-prints*, arXiv:2106.11349 (June 2021), arXiv:2106.11349.
[67] C. J. Leininger and A. W. Reid. "A combination theorem for Veech subgroups of the mapping class group". In: *Geom. Funct. Anal.* 16.2 (2006), pp. 403–436.
[68] Christopher J. Leininger. "Accidental parabolics in the mapping class group". In: *Proc. Amer. Math. Soc.* 137.3 (2009), pp. 1153–1160.
[69] Christopher Loa. "Free products of abelian groups in mapping class groups". In: *arXiv e-prints*, arXiv:2103.05144 (Mar. 2021), arXiv:2103.05144.

[70] Kathryn Mann and Jason Fox Manning. "Stability for hyperbolic groups acting on boundary spheres". English. In: *Forum Math. Sigma* 11 (2023). Id/No e83, p. 25.
[71] Kathryn Mann, Jason Fox Manning, and Theodore Weisman. *Stability of hyperbolic groups acting on their boundaries*. Preprint, arXiv:2206.14914. 2022.
[72] Kathryn Mann, Jason Fox Manning, and Theodore Weisman. "Topological stability of relatively hyperbolic groups acting on their boundaries". In: *arXiv e-prints*, arXiv:2402.06144 (Feb. 2024), arXiv:2402.06144.
[73] Howard Masur. "Transitivity properties of the horocyclic and geodesic flows on moduli space". In: *J. Analyse Math.* 39 (1981), pp. 1–10.
[74] Howard A. Masur and Yair N. Minsky. "Geometry of the complex of curves. I. Hyperbolicity". In: *Invent. Math.* 138.1 (1999), pp. 103–149.
[75] Howard A. Masur and Yair N. Minsky. "Geometry of the complex of curves. II. Hierarchical structure". In: *Geom. Funct. Anal.* 10.4 (2000), pp. 902–974.
[76] Lee Mosher. "Problems in the geometry of surface group extensions". In: *Problems on mapping class groups and related topics*. Vol. 74. Proc. Sympos. Pure Math. Amer. Math. Soc., Providence, RI, 2006, pp. 245–256.
[77] Hossein Namazi and Juan Souto. "Non-realizability and ending laminations: proof of the density conjecture". In: *Acta Math.* 209.2 (2012), pp. 323–395.
[78] Thang Nguyen and Theodore Weisman. "Stability for boundary actions of cocompact lattices in Euclidean buildings". In: *arXiv e-prints*, arXiv:2607.14668 (July 2026), arXiv:2607.14668.
[79] Graham Niblo and Lawrence Reeves. "Coxeter groups act on CAT(0) cube complexes". In: *J. Group Theory* 6.3 (2003), pp. 399–413.
[80] Ken'ichi Ohshika. "Realising end invariants by limits of minimally parabolic, geometrically finite groups". In: *Geom. Topol.* 15.2 (2011), pp. 827–890.
[81] Yann Ollivier and Daniel T. Wise. "Cubulating random groups at density less than 1/6". In: *Trans. Amer. Math. Soc.* 363.9 (2011), pp. 4701–4733.
[82] Denis V. Osin. "Peripheral Fillings of Relatively Hyperbolic Groups". In: *Inventiones Mathematicae* 167.2 (2007), pp. 295–326.
[83] Kasra Rafi. "Hyperbolicity in Teichmüller space". In: *Geom. Topol.* 18.5 (2014), pp. 3025–3053.
[84] Jacob Russell, Davide Spriano, and Hung Cong Tran. "The local-to-global property for Morse quasi-geodesics". In: *Math. Z.* 300.2 (2022), pp. 1557–1602.
[85] Nick Salter and Bena Tshishiku. "Surface bundles in topology, algebraic geometry, and group theory". In: *Notices Amer. Math. Soc.* 67.2 (2020), pp. 146–154.
[86] Richard Evan Schwartz. *Spherical CR Geometry and Dehn Surgery (AM-165)*. Princeton University Press, 2007.
[87] Dennis Sullivan. "Quasiconformal homeomorphisms and dynamics II: Structural stability implies hyperbolicity for Kleinian groups". In: *Acta Mathematica* 155.1 (1985), pp. 243–260.
[88] Dennis Sullivan. "Quasiconformal homeomorphisms and dynamics. II. Structural stability implies hyperbolicity for Kleinian groups". In: *Acta Math.* 155.3-4 (1985), pp. 243–260.
[89] Alec Traaseth and Theodore Weisman. "Combination theorems for geometrically finite convergence groups". In: *Algebr. Geom. Topol.* 26.6 (2026), pp. 2079–2122.
[90] Konstantinos Tsouvalas and Theodore Weisman. "Singular value gap estimates for free products of semigroups". In: *arXiv preprint arXiv:2409.20330* (2024).
[91] Brian Udall. "Combinations of parabolically geometrically finite groups and their geometry". In: *arXiv e-prints*, arXiv:2307.06468 (July 2023), arXiv:2307.06468.
[92] E. B. Vinberg. "Discrete Linear Groups That Are Generated by Reflections". In: *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* 35 (1971), pp. 1072–1112.
[93] Tianqi Wang. "Notions of Anosov representation of relatively hyperbolic groups". In: *arXiv e-prints*, arXiv:2309.15636 (Sept. 2023), arXiv:2309.15636.
[94] Theodore Weisman. "An extended definition of Anosov representation for relatively hyperbolic groups". In: *J. Topol.* 19.2 (2026), Paper No. e70070, 71.
[95] Theodore Weisman. "Dehn filling in semisimple Lie groups". In: *arXiv e-prints*, arXiv:2502.17592 (Feb. 2025), arXiv:2502.17592.

[96] Theodore Weisman. "Dynamical properties of convex cocompact actions in projective space". In: *Journal of Topology* 16.3 (2023), pp. 990–1047.

[97] Theodore Weisman. "Examples of extended geometrically finite representations". In: *arXiv e-prints*, arXiv:2311.18653 (Nov. 2023), arXiv:2311.18653.

[98] Anna Wienhard. "An invitation to higher Teichmüller theory". In: *Proceedings of the International Congress of Mathematicians: Rio de Janeiro 2018*. World Scientific. 2018, pp. 1013–1039.

[99] Daniel T. Wise. "Cubulating small cancellation groups". In: *Geom. Funct. Anal.* 14.1 (2004), pp. 150–214.

[100] Daniel T. Wise. "The cubical route to understanding groups". In: *Proceedings of the International Congress of Mathematicians—Seoul 2014. Vol. II*. Kyung Moon Sa, Seoul, 2014, pp. 1075–1099.

[101] Daniel T. Wise. *The structure of groups with a quasiconvex hierarchy*. Vol. 209. Annals of Mathematics Studies. Princeton University Press, Princeton, NJ, 2021, pp. x+357.

[102] Asli Yaman. "A topological characterisation of relatively hyperbolic groups". In: *J. Reine Angew. Math.* 566 (2004), pp. 41–89.

[103] Feng Zhu. "Relatively dominated representations". In: *Ann. Inst. Fourier (Grenoble)* 71.5 (2021), pp. 2169–2235.

[104] Feng Zhu and Andrew Zimmer. "Relatively Anosov representations via flows I: theory". In: *arXiv e-prints*, arXiv:2207.14737 (July 2022), arXiv:2207.14737.

[105] Andrew Zimmer. "Projective Anosov representations, convex cocompact actions, and rigidity". In: *J. Differential Geom.* 119.3 (2021), pp. 513–586.